



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018**

CANDIDATE
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CENTRE
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H2 BIOLOGY

Paper 2 Structured Questions

9744/02

**Friday 24 August 2018
2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, Centre number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use					
Q1	/ 8	Q5	/ 10	Q9	/ 12
Q2	/ 11	Q6	/ 10	Q10	/ 8
Q3	/ 11	Q7	/ 5	Q11	/ 8
Q4	/ 7	Q8	/ 10		
Total					/ 100

This document consists of **27** printed pages and **1** blank page.

Answer **all** the questions in this section.

1 Plants vary greatly in terms of size.

(a) Explain whether the cell theory is applicable to plants. [2]

1. Applicable.
2. Plants are living organisms, which are composed of (many, different plant) cells,
3. which are basic/ smallest unit of life.
4. All plant cells come from pre-existing plant cells via cell division (e.g. mitosis or meiosis).

Sugar molecules enter cells through transport proteins.

(b) Explain why transport proteins are required for the movement of sugar molecules, such as glucose and fructose, into cells. [2]

1. Glucose and fructose are polar molecules.
2. They are unable to cross
3. the hydrophobic core of the phospholipid bilayer.
4. Transport proteins shield them from hydrophobic core of plasma membrane (e.g. channel proteins provide a hydrophilic channel for their movement across the membrane).

Some plant cells convert fructose and glucose into sucrose for transport from the leaves to the roots. Sucrose is moved into phloem sieve tubes as shown in Fig. 1.1.

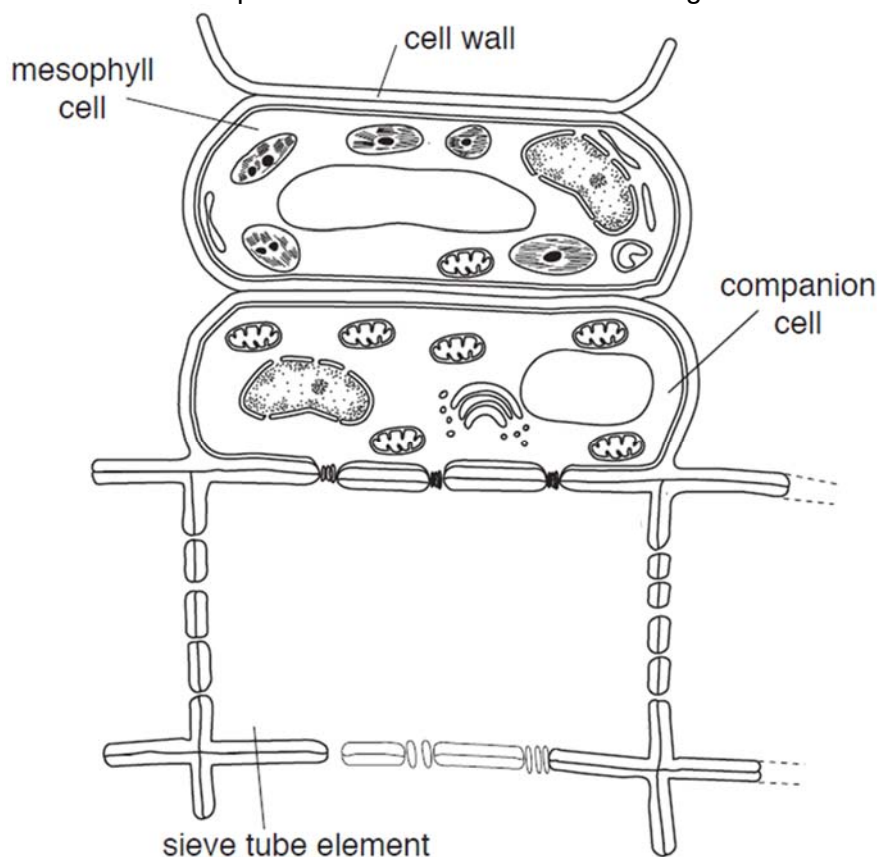


Fig. 1.1

Each cell has a specialized function.

(c) With reference to Fig. 1.1 and the information provided, state **one** difference between a mesophyll cell and companion cell. [1]

1. Companion cells (6 mitochondria) have more mitochondria than mesophyll cells (1 mitochondrion). [1]

OR

Mesophyll cells (5 chloroplasts) have chloroplasts whereas companion cells have none. [1]

Fig. 1.2 shows how sucrose is transported into the companion cell from the mesophyll cell.

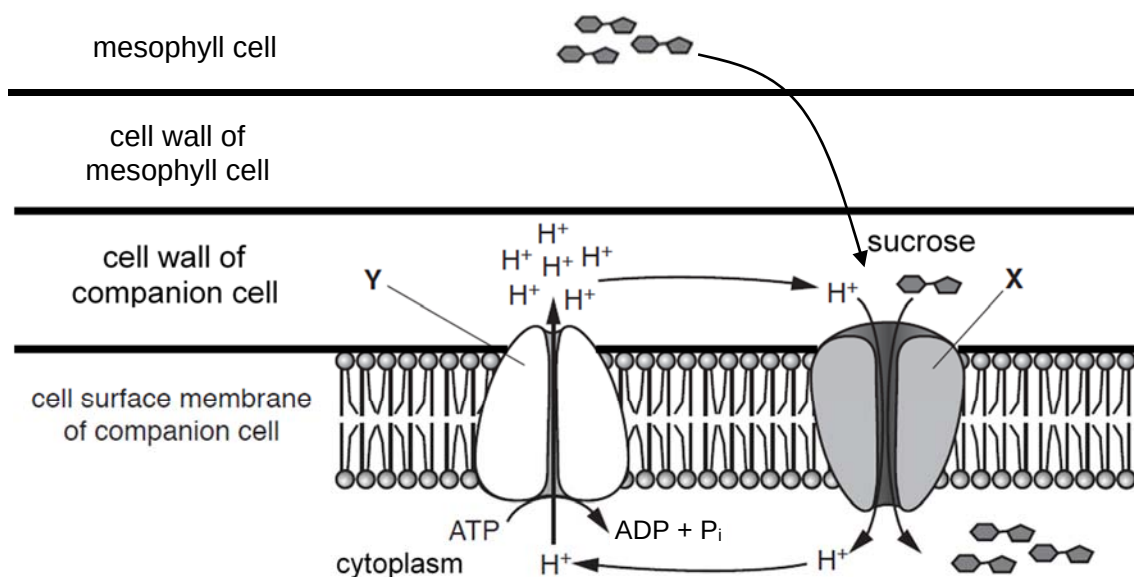


Fig. 1.2

(d) Using the information in Fig. 1.1 and Fig. 1.2, explain how sucrose moves into the companion cell. [3]

1. Sucrose diffuses from mesophyll cell to the cell wall of companion cell.
2. Protons are actively pumped out from the cytoplasm of companion cell into its cell wall through carrier protein Y via active transport (hydrolysis of ATP). [1]
[Reject: Diffuse]
3. Protons then diffuses from the cell wall of companion cells into the companion cell through transport protein X (cotransporter) via facilitated diffusion [1]
4. which is coupled with the transport of sucrose
5. against the sucrose concentration gradient.

[Total: 8]

- 2 The yeast, *Saccharomyces cerevisiae*, is a single-celled, eukaryotic organism that is often used in the laboratory.

When yeast is mixed with a glucose solution, the yeast absorbs the glucose. Each molecule of glucose is then broken down into pyruvate molecules in exactly the same way as in any other eukaryotic organism.

- (a) Outline the breakdown of glucose to pyruvate in this stage. [2]

Respiration Lecture Notes p.9, 10

1. **Glucose is broken to pyruvate during glycolysis.**
2. **Glucose is first phosphorylated to glucose-6-phosphate**
3. **which is (isomerized to fructose-6-phosphate and then phosphorylated to) converted to fructose-1,6-bisphosphate**
4. **before being cleaved/ broken down into 2 three-carbon sugars (OR glyceraldehyde-3-phosphate and dihydroxyacetone phosphate),**
5. **which is then oxidised/ converted to form 2 molecules of pyruvate.**

Yeast cells sometimes carry out anaerobic respiration. Fig. 2.1 outlines the process of anaerobic respiration in yeast cells.

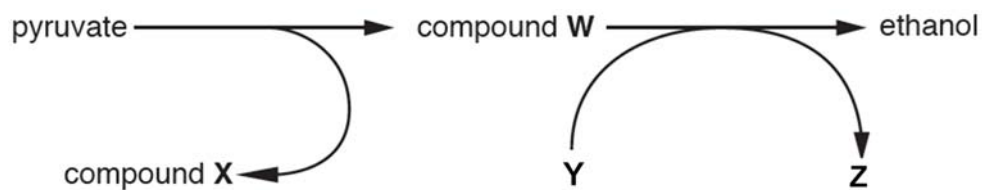


Fig. 2.1

- (b) (i) Identify molecule **Z**. [1]
NAD or NAD⁺
- (ii) State why molecule **Y** is converted to **Z**. [1]
Respiration Lecture Notes p.26
 1. **Regenerate NAD**
 2. **required for glycolysis to continue.**

[Accept: Reduce compound W (ethanal) to ethanol]

Yeasts are often used in bread-making. The bread dough is kneaded to introduce and trap air so that the yeasts in the dough can respire aerobically. Besides carbon dioxide that is released during respiration, the evaporation of water or ethanol released during respiration also causes the dough to rise.

Table 2.1 shows the differences in the height of dough that was placed at different locations, after the dough was kneaded.

Table 2.1

Time / min	Height of dough / cm		
	Fridge	Room temperature	Next to window (hot day)
0	2.5	2.5	2.5
20	2.5	2.9	3.3
40	2.7	3.7	4.0
60	2.9	3.9	4.7
80	3.0	4.0	5.2
100	3.0	4.0	5.8
120	3.0	4.0	6.0

(c) (i) Account for the difference in the overall increase in the height of dough that was placed in the fridge with that placed next to the window. [4]

1. The height of the dough when placed in the fridge (F) increases from 2.5 at 0 min to 3.0cm at 120 min is LOWER than that when placed next to window (W) which increases from 2.5 to 6.0 cm. [1]
2. The temperature of the dough in F is lower than that of W.
3. Hence, the kinetic energy of respiratory enzymes and substrates is lower.
[Accept: Enzymes are inactivated]
4. The frequency of effective collisions between enzymes and substrates is lower
5. hence the rate of formation of enzyme-substrate complexes is lower.
6. The rate of respiratory enzyme activity / rate of respiration is lower.
7. and less carbon dioxide are released and less evaporation of water or ethanol, which causes the dough to rise less.

(ii) Suggest why the increase in the height of dough that was placed at room temperature was higher between 0 and 40 minutes than between 40 minutes and 60 minutes. [2]

1. The height of dough increases from 2.5 to 3.7cm between 0 and 40 min is HIGHER than 3.7 to 3.9cm between 40 and 60 minutes. [1]

WITH

2. There are more oxygen between 0 and 40 min, hence the yeast undergoes aerobic respiration which releases more molecules of CO₂ (6 molecules of CO₂)

and 6 molecules of H₂O) than anaerobic respiration (2 molecules of CO₂ and 2 molecules of ethanol) from 40 minutes and 60 minutes. [1]

OR

There are more respiratory substrates at the start between 0 and 40 min, the rate of formation of enzyme-substrate complexes is higher, hence the rate of respiration is higher than between 40 minutes and 60 minutes. [1]

OR

From 40 minutes and 60 minutes, the yeast undergoes anaerobic respiration and high concentration of ethanol produced is toxic and kills the yeast. [1]

(iii) Suggest why the height of the dough that was placed at room temperature ceases to increase after 80 minutes. [1]

1. The high concentration of ethanol produced is toxic and kills the yeast. [1]

[Total: 11]

- 3 In maize plants, a gene locus for leaf colour and a gene locus for cob colour were studied.

A pure breeding maize plant with bronze leaves and brown cob was crossed with a pure breeding maize plant with green leaves and yellow cobs to produce F1 phenotypes.

All the F1 plants had bronze leaves and brown cobs.

- (a) Define the term *locus*. [1]

[Genetics I Lecture Notes p.4](#)

1. [Fixed position/ location](#)
2. [on a particular chromosome](#)

A test cross was conducted for these two loci using the F1 plant. Table 3.1 shows the results of this cross.

Table 3.1

Phenotype	Observed number (O)
bronze leaves and brown cobs	44
bronze leaves and yellow cobs	6
green leaves and brown cobs	7
green leaves and yellow cobs	43
Total	100

- (b) (i) State the phenotype of the test cross plant. [1]

[Plant for green leaves and yellow cobs](#)

(ii) Draw a genetic diagram to explain the results of this test cross. [4]

Use the symbols:

L bronze leaves; **I** green leaves; **B** brown cobs; **b** yellow cobs

TESTCROSS

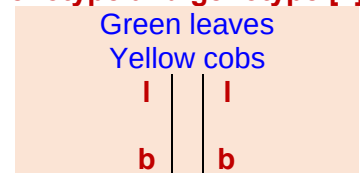
Testcross parental phenotype:

Testcross parental genotype:



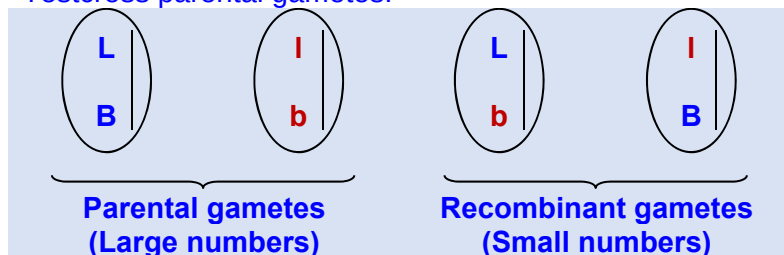
Parental phenotype and genotype [1]

×



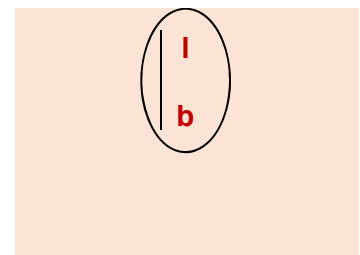
×

Testcross parental gametes:



Parental gametes [1]

×



Punnett square, offspring genotype and phenotype, phenotypic ratio [2]

Offspring genotype:

Offspring phenotype:

Offspring phenotypic ratio:

Observed numbers:

L	I
B	b
Bronze leaves	Green leaves
Brown cobs	Yellow cobs
1	1
44	43

Large numbers
Non-recombinant phenotype

L	I
b	b
Bronze leaves	Green leaves
Yellow cobs	Brown cobs
1	1
6	7

Small numbers
Recombinant phenotype

Another type of maize plant produced a total of 381 grains, 216 purple and smooth, 79 purple and shrunken, 65 yellow and smooth and 21 yellow and shrunken.

A chi-squared test was carried out to test the significance of the differences between the observed and expected results.

Table 3.2 shows some calculations to obtain the chi-squared value.

Table 3.2

grain phenotype	observed number	expected number	$\frac{[\text{obs no.} - \text{exp no.}]^2}{\text{exp no.}}$
purple and smooth	216	$381 \times 9/16 = 214$	$4/214 = 0.019$
purple and shrunken	79	$381 \times 3/16 = 71$	$64/71 = 0.901$
yellow and smooth	65	$381 \times 3/16 = 71$	$36/71 = 0.507$
yellow and shrunken	21		
Total number	381	χ^2 value	

(c) Complete the missing spaces in Table 3.2.

[2]

grain phenotype	observed number	expected number	$\frac{[\text{obs no.} - \text{exp no.}]^2}{\text{exp no.}}$
purple and smooth	216	$381 \times 9/16 = 214$	$4/214 = 0.019$
purple and shrunken	79	$381 \times 3/16 = 71$	$64/71 = 0.901$
yellow and smooth	65	$381 \times 3/16 = 71$	$36/71 = 0.507$
yellow and shrunken	21	$381 \times 1/16 = 24$	$9/24 = 0.375$
Total number	381	χ^2 value	1.80

[1 for each row, allow ecf]

Table 3.3 shows some critical values for chi-squared test at different probability levels.

Table 3.3

Degrees of freedom	Probability, p						
	0.50	0.10	0.05	0.02	0.01	0.005	0.001
1	0.46	2.71	3.84	5.41	6.64	7.88	10.83
2	1.39	4.61	5.99	7.82	9.21	10.60	13.82
3	2.37	6.25	7.82	9.84	11.35	12.84	16.27
4	3.36	7.78	9.49	11.67	13.28	14.86	18.47
5	4.35	9.24	11.07	13.33	15.09	16.75	20.51

- (d) (i) Describe how the degrees of freedom was determined; [1]
 1. The degree of freedom is total number of categories/ phenotypes (i.e. 4)
 2. minus one.
- (ii) State the conclusion from the χ^2 value calculated in (c). [2]
 1. For 3 degrees of freedom,
 2. the calculated χ^2 value of 1.80 is less than the critical χ^2 value of 7.82.
 3. The probability that the difference is due to chance is greater than 0.05.
 4. The difference between observed numbers and expected ratio is not statistically significant and it is due to chance.

[Total: 11]

✂ End of Part 1 ✂

2018 PRELIMINARY EXAMINATION

H2 BIOLOGY PAPER 2 [PART 2]:

Structured Questions

Name: _____

Civics Group: _____/17

For Examiner's Use

Q4	/ 7
Q5	/ 10
Q6	/ 10
Q7	/ 5

- 4 Fig. 4.1 shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

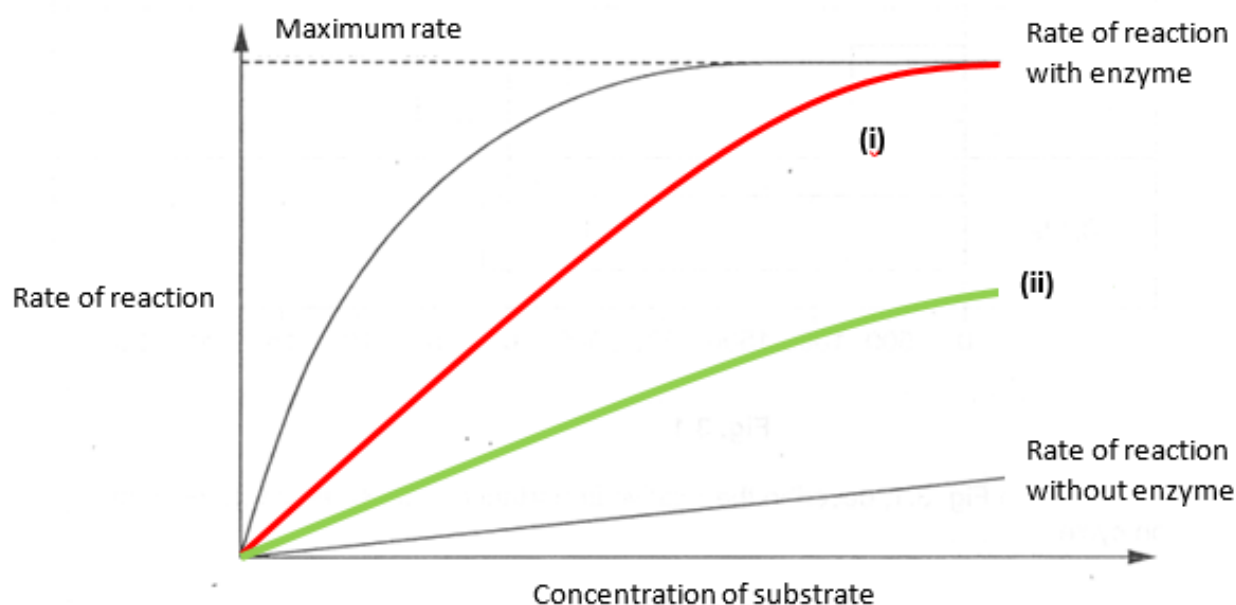


Fig. 4.1

- (a) On Fig. 4.1, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction upon the addition of

- (i) a competitive inhibitor;
(ii) a non-competitive inhibitor.

[2]

- (b) Explain the effect of a competitive inhibitor on the rate of enzyme activity. [3]

- Shape of inhibitor is similar in shape of substrate
- Shape of inhibitor is complementary to the shape of active site
- Competitive inhibitors compete with the substrate molecules for the active site and bind at the active site of the enzyme

4. blocking / prevents substrate molecules from binding to active site,
5. reducing
 - i. number of enzyme-substrate complex formed per unit time
or
 - ii. rate of enzyme-substrate complex formation
6. thus decreasing rate of enzyme activity

(c) State **two** differences between a competitive inhibitor and a non-competitive inhibitor. [2]

Point of comparison	competitive inhibitor	non-competitive inhibitor
Binding site	Enzyme active site	region other than its active site / allosteric binding site
Structure	Structurally similar to substrate.	Structure not similar to substrate.
Overcoming its effects	Effects can be overcome by increasing substrate concentration	Effects cannot be overcome by increasing substrate concentration

[Total: 7]

- 5 Fig. 5.1 shows a stage in the mitotic cell cycle in an animal cell.

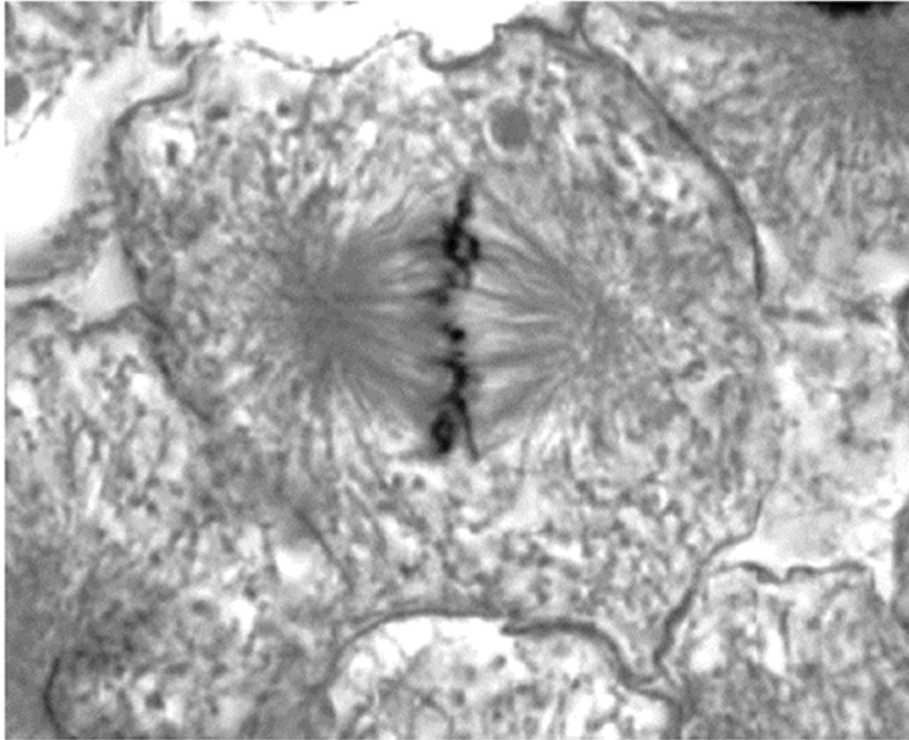


Fig. 5.1

- (a) With reference of Fig. 5.1,

- (i) identify the stage of mitosis; [1]

Metaphase

- (ii) state two features which are characteristic of this stage. [2]

1. Chromosomes line up at the equator of the cell/ metaphase plate. [1]
2. Centromere/kinetochore attached to spindle fibres/microtubules from the centrioles. [1]
3. Centrioles reach/located at poles of the cell. [1]

- (b) Distinguish between the terms haploid and diploid. [2]

1. Haploid refers to only one set of chromosome being present in a cell.
 2. Whereas diploid refers to cell having 2 sets of chromosomes.
- OR
3. Haploid condition consists of one member of each pair of homologous chromosome present.
 4. Diploid condition consists of 2 sets of chromosomes, one set derived from each parent.

(c) Explain the importance of mitosis in organisms. [3]

1. maintains / same, genetic stability / number of chromosomes/ two sets of chromosomes / diploid / $2n$ /
2. produces daughter cells that are genetically identical
3. replacement of cells ;
4. repair of tissue ;
5. growth / increase in cell numbers ;
6. asexual reproduction;

(d) In many multicellular organisms, such as mammals, the time taken for the mitotic cell cycle varies considerably between different tissues, but is very carefully controlled in each cell.

Suggest the importance of this control in mammals. [2]

1. Prevent tumour/ cancer formation due to uncontrolled cell division. [1]
2. Only cells that are needed / functions are needed will be produced [1]
3. Allows for coordination of growth / limiting growth ; [1]

[Total: 10]

6 *Staphylococcus aureus* is a bacterium that is resistant to most types of antibiotics, such as penicillin.

In a study to understand how bacteria gain antibiotic resistance, a strain of *E. coli* with no known antibiotic resistance was mixed with heat-killed *S. aureus* for 24 hours.

E. coli was then grown on Petri dish containing penicillin and the number of *E. coli* colonies were counted.

For the control, *E. coli* without *S. aureus* was grown on another Petri dish and the number of colonies were counted.

Fig 6.1 shows the result of the experiment.

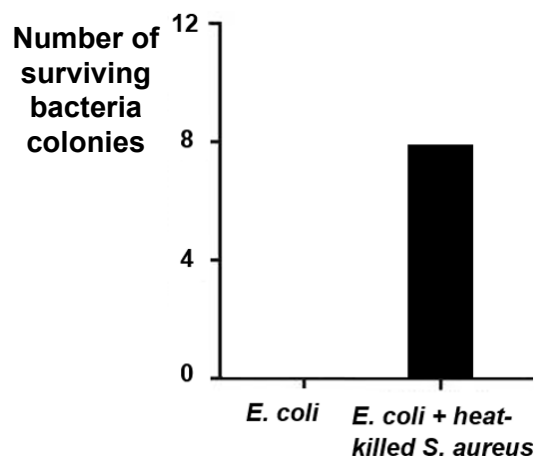


Fig. 6.1

- (a) Identify the process that allows *E. coli* to become antibiotic resistant. [1]

Transformation

- (b) With reference to Fig. 6.1,

- (i) describe the results observed; [1]

- **After mixing heat-killed *S. aureus* and *E. coli*, the number of surviving *E. coli* colonies increased from 0 to 8 colonies/ml**

- (ii) Explain the results observed in (b)(i). [2]

1. Heat - killed *S. aureus* released a piece of DNA that codes for penicillin resistance.

2. When mixed with the original *E. coli*, competent *E. coli* bacteria would take in this piece of DNA containing penicillin resistance.

3. The DNA is then integrated into the bacterial chromosome of the *E. coli*

4. and expressed, making it resistant to penicillin.

Fig. 6.2 shows another process by which antibiotic resistance genes can be passed from bacterium A to B.

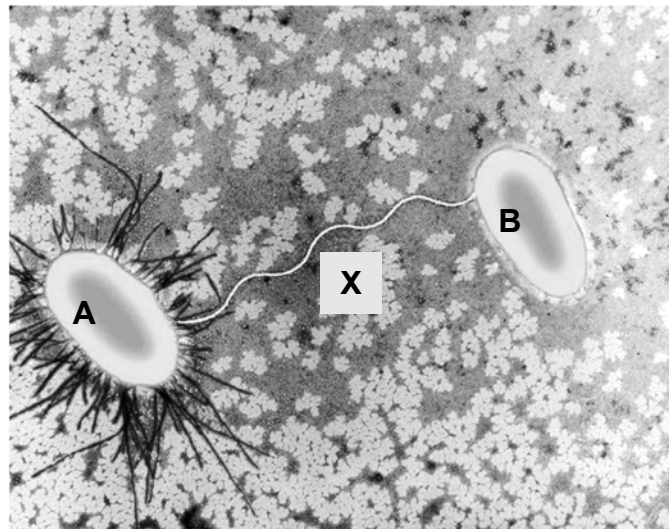


Fig. 6.2

- (c) With reference to the process shown in Fig. 6.2,

- (i) identify the nature of each bacterium; [2]

Bacterium A : **F+ (REJECT DONOR)**

Bacterium B : **F- (REJECT RECIPIENT)**

- (ii) identify structure X.

X : Conjugation bridge (REJECT SEX PILUS)

[1]

- (iii) Distinguish between the process of genetic recombination stated in (a) and in Fig. 6.2. [3]

	Transformation	Conjugation
Source of foreign DNA	<u>Naked foreign DNA from environment/ <i>S. Aureus</i></u>	<u>F plasmid in a F⁺ cell</u>
Type of bacterial cell needed for the process	<u>Competent bacteria cell</u>	<u>F⁺ cell</u>
Contact between cells	<u>Not required</u>	<u>required</u>
Description of Foreign DNA	<u>Any pieces of DNA.</u>	▪ <u>F plasmid</u>
Type of genes transferred	<u>Random</u>	▪ Only genes found on the F plasmid
Outcome of recipient bacterial cell	<u>Does not become F⁺ cell</u>	▪ Becomes a F⁺ cell ▪

[Total: 10]

- 7 In an investigation to study genetic variation, DNA was obtained from four varieties of the same invertebrate species.

The following technique was used:

- DNA was digested using a number of different restriction enzymes to obtain different fragments
- The fragments were separated by gel electrophoresis
- RNA probes were used to select DNA fragments with specific sequences

(a) Explain how RNA probes, used in this technique, select fragments of DNA. [2]

1. RNA probes are single stranded and [1]
2. are complementary to a target sequence on a DNA fragment will hybridize with the probe. [1]

Fig. 7.1 shows the results after the RNA probes have bound to the selected DNA fragments.

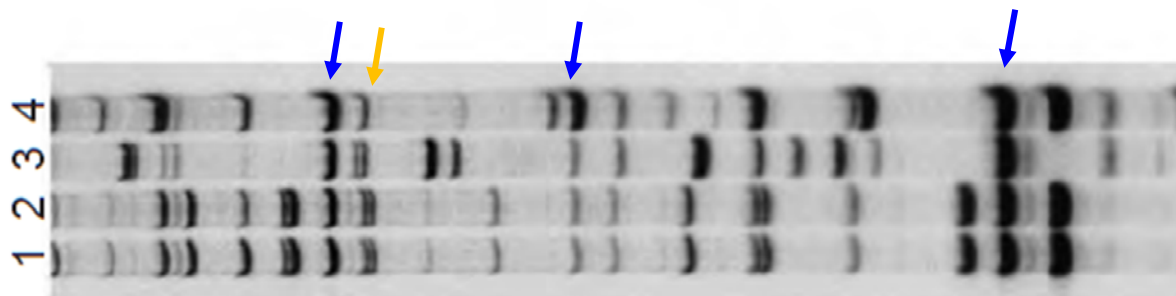


Fig. 7.1

Any row that has all 4 bands

(b) On Fig. 7.1,

- (i) draw an arrow (→), indicate **one** DNA fragment found in all four varieties where the RNA probe has bound; [1]
- (ii) Identify the varieties which have the same genetic fingerprint and explain your answer.

1. **varieties 1 and 2;**
2. **they have, a very similar, pattern of DNA bands / DNA banding pattern**

[Total: 5]

⌂ End of Part 2 ↻

- 8 Table 8.1 shows some of the common fatty acids and their melting points.

Table 8.1

Symbol (number of carbon atoms : number of double bonds)	Common Name	Melting point (°C)
<i>Saturated fatty acids</i>		
12 : 0	Lauric acid	44.2
14 : 0	Myristic acid	52
16 : 0	Palmitic acid	63.1
18 : 0	Stearic acid	69.6
20 : 0	Arachidic acid	75.4
22 : 0	Behenic acid	81
<i>Unsaturated fatty acids</i>		
16 : 1	Palmitoleic acid	-0.5
18 : 1	Oleic acid	13.4
18 : 2	Linoleic acid	-9
18 : 3	α -linolenic acid	-17
20 : 4	Arachnidonic acid	-49.5

- (a) Arachidonic acid is a polyunsaturated fatty acid. Explain the term *polyunsaturated fatty acid*.
[1]

- A fatty acid with many C=C double bonds.
Reject : many kinks

- (b) With reference to Table 8.1,

- (i) describe the effect of increasing number of carbon atoms in saturated fatty acids on the melting point; [3]

1. As the number of carbon atoms increased 12 to 22, the melting point increased from 44.2 to 81 °C.
2. An initial increase of every 2 carbon atoms from 12 to 18 leads to a sharp increase in the melting point from 44.2 to 69.6 °C.
3. Further increase of every 2 carbon atoms from 18 to 22 lead to a lesser increase in melting point from 69.6 to 81 °C.
4. As the number of carbon atoms increases, the melting point increases.

- (ii) describe the effect of the presence of double bonds in fatty acids on the melting point; [1]

1. As the number of double bonds increases, the melting point decreases.

2. As the number of double bonds increased from 1 (in oleic acid) to 3 (in α -linolenic acid), the melting point decreased from 13.4 to -17 °C.

(iii) explain the trend described in **b(ii)**. [4]

1. Presence of double bonds results in the fatty acid molecules being bent/ kinked.
2. This means that the molecules cannot be closely packed together / less contact between molecules,
3. resulting in weaker hydrophobic interactions.
4. Therefore, less energy required to overcome the hydrophobic interactions / separate the fatty acid molecules during melting, resulting in the decrease in melting point.

(c) Suggest where polyunsaturated fatty acids are usually found in nature. [1]

- Vegetable oils
- Nuts
- Cold water fish

[Total: 10]

- 9 Table 9.1 provides statements regarding the bonds found in four biological molecules.

Table 9.1

statement	protein	DNA	messenger RNA	cellulose
hydrogen bonds stabilise the molecule	✓	✓	×	✓
subunits are joined by peptide bonds	✓	×	×	×

- (a) Complete Table 9.1 by indicating with a tick (✓) or a cross (✗) whether the statements apply to proteins, DNA, messenger RNA and cellulose.

You should put a tick or a cross in each box of the table.

[2]

- (b) Telomeres are parts of chromosomes. Describe the function of telomeres. [4]

1a. Protects the organism's genes from being lost with each cycle of DNA replication / genetic material / DNA

1b. due to gap at the 5' end of each replicated DNA strand / DNA shortened

2a. Protect chromosomal ends from degradation

2b. by binding proteins to form telomere caps.

3a. Prevents ends of chromosomes attaching to each other

3a. prevents apoptosis / prevent chromosomal ends from activating cell's system for monitoring DNA damage.

4a. Enables lengthening of telomeres by

4b. providing a recognition site for the enzyme telomerase.

- (c) A piece of mRNA is 660 nucleotides long but the DNA coding strand from which it was transcribed is 870 nucleotides long.

- (i) Explain this difference in number of nucleotides. [1]

- Introns present in DNA
- Introns absent in mRNA

OR

- introns removed by RNA splicing

- (ii) What is the maximum number of amino acids in the protein translated from this piece of mRNA? Explain your answer. [2]

Number of amino acids 220 OR 219

Explanation

1. 3 bases code for 1 amino acids

- (d) Identify **one** other process that leads to the formation of mature mRNA and state its function. [2]

1. Addition of 5' cap

[Significance]

2. facilitate the binding of Translation Initiation Factors and small ribosomal subunit for translation to occur.
OR
2. facilitate the export of mature mRNA from nucleus to cytoplasm for translation
OR
2. protect the mature mRNA from degradation by RNase in the cytoplasm

OR

1. Addition of 3' poly-A tail or 3' polyadenylation

[Significance]

2. facilitate the export of mature mRNA from nucleus to cytoplasm for translation
OR
3. protect the mature mRNA from degradation by RNase in the cytoplasm

- (e) Describe **one** difference between the structure of mRNA and tRNA. [1]

Any one:

1. mRNA has no base-pairing within its structure while tRNA has base-pairing between regions to fold back on itself.
2. mRNA has 3' poly-A tail while tRNA has 3' CCA end.
3. mRNA does not have hydrogen bonds different regions of the single strand while tRNA has hydrogen bonds at different regions which cause it to fold back on itself.
4. mRNA is linear while tRNA cloverleaf shape;
5. mRNA has no binding site for amino acids while tRNA has.
6. mRNA longer/larger/more nucleotides than tRNA
7. Mrna different for each gene/many kinds, only few/20/64 kinds of tRNA;

[Total: 12]

- 10 Unlike eukaryotes, prokaryotes have different mechanisms for controlling gene expression. Fig. 10.1 shows the Jacob and Monod model of gene expression in the *lac* operon of *Escherichia coli*.

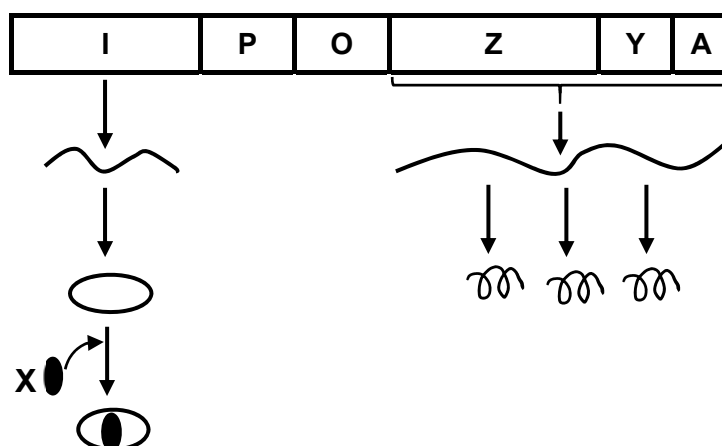


Fig. 10.1

(a) With reference to Fig. 10.1 identify,

- (i) identify molecule X; [1]
 - **allolactose**
- (ii) identify region I. [1]
 - **Regulatory gene / lac I gene / Repressor gene**

The regulation of *lac* operon in *E. coli* was investigated and wild-type *E. coli* were cultured in two different agar media.

X-gal was added to both agar media. It is a colourless substance that is converted to a blue compound by the enzyme, β -galactosidase.

Table 10.1 shows results of the investigation.

Table 10.1

Type of agar medium	Colour of colony
X-gal, lactose and no glucose	blue
X-gal, lactose and glucose	white

(b) Apart from the presence of an inactive *lac* repressor, explain the appearance of the colony when wild-type *E. coli* was cultured in lactose and X-gal without glucose. [3]

1. When glucose is absent, there is an **increase / high concentration of cAMP**.
2. **cAMP binds to** the allosteric site on **CAP (catabolite activator protein)**.

3. The CAP in active shape and binds to a CAP-binding site at the upstream end of the *lac* promoter.
4. The attachment of CAP bends the DNA, which makes it easier for RNA polymerase to bind to the promoter.
5. The *Lac* operon is transcribed.
6. High amount of β -galactosidase is produced and breaks down the colourless X-gal to a blue compound.

In a separate experiment, scientists fused the *trp* operon with the *lac* operon as shown in Fig.10.2.

The *trp-lac* fusion operon was then inserted into a bacterial cell to replace the separate *trp* operon and *lac* operon such that the transformed bacterium only has the *trp-lac* fusion operon.

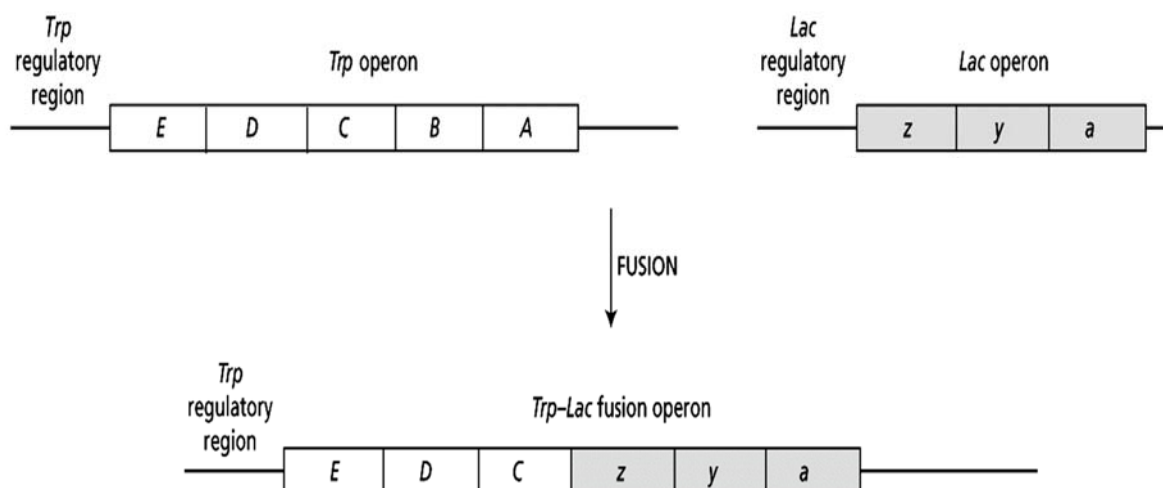


Fig. 10.2

(c) State and explain the conditions which must be present in order for β -galactosidase to be formed in the transformed bacterium with *trp-lac* fusion operon. [3]

1. Tryptophan of very low levels / absent from the medium.
2. *trp* repressor remains in the inactive conformation and
3. does not bind to operator and
4. RNA polymerase binds to promoter
5. Transcription of both *trp* and *lac* structural genes takes place.
6. Because the fused operon, *lac* operon under control of *trp* regulatory region.
7. Tryptophan must be very low levels / absent from the medium.
8. In the fused operon, the *lac* operon now under the control of the *trp* regulatory region.

9. If corepressor / tryptophan does not bind to the trp repressor,
 10. trp repressor remains in the inactive conformation and does not bind to operator and
 11. RNA polymerase binds to promoter
 12. Transcription of structural genes and translation take place.
- no longer dependent on glucose or lactose levels.

[Total: 8]

11 Relationships between different primates can be found by comparing their proteins and DNA.

The proteins of different species can be compared using immunological techniques.

The protein albumin obtained from a human was injected into a rabbit. The rabbit produced antibodies against the human albumin.

These antibodies were extracted from the rabbit and then added to samples of albumin obtained from four different animal species. Precipitation occurs when antibodies bind to albumin. The amount of precipitate produced in each sample was then measured and shown in Table 11.1.

Table 11.1

Species from which albumin was obtained	Amount of precipitate / arbitrary units
Rat	23
Chimpanzee	96
Marmoset	65
Trout	11

(a) Comment on what the results suggest about the evolutionary relationship between humans and the other species? [2]

1. Human is most closely related to chimpanzee, followed by marmoset, rat and least closely related to trout;
2. Amount of precipitate formed with chimpanzee is the highest at 96a.u., followed by marmoset (65a.u), than rat (23a.u.) and lowest is trout at 11a.u.

Scientists also used DNA hybridisation to determine the evolutionary relationships between five species of primate.

The separation temperature is the temperature at which a molecule of double-stranded DNA separates into two single strands.

The scientists first recorded the mean separation temperature of DNA in which both strands were from the same species.

The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 11.2.

Table 11.2

Primate	Mean decrease in separation temperature / °C				
	Human	Chimpanzee	Gorilla	Orang-utan	Gibbon
Human					
Chimpanzee	1.7				
Gorilla	2.3	2.3			
Orang-utan	3.6	3.6	3.5		
Gibbon	4.8	4.8	4.7	4.9	

- (b) When the scientists first recorded the mean separation temperature of DNA in which both strands were from the same species, differences in the separation temperature was observed. Suggest why this is so. [1]

- Individuals within same species have different alleles / different base sequences / (different) mutations / introns ;

- (c) With reference to Table 11.2,

- (i) explain if the data suggests that gibbons are most distantly related to humans; [2]

- Yes
- There is largest / highest decrease in separation temperature of 4.8 °C compared to the other species
- This means that there are fewer complementary bases between the DNA strand from human compared to gibbon.
- Fewer hydrogen bonds present, less energy needed to separate the strands.

- (ii) The scientists assumed that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of these primates separated.

Gorillas are thought to have separated from orang-utans 20 million years ago. Use this information to calculate how long ago the evolutionary lines of humans and chimpanzees separated.

Show your working. [3]

- Answer in the range of 9.69 to 9.714286**

Working

- 3.5 °C represents 20 million years**
- For 1 °C = $20,000,000 \div 3.5 = 5.7$ million years or 5,714 286 million years**
- Humans and chimpanzees would have separated = $1.7 \times 5.7 = 9.69$ million years [1/2]**

_____ million years

[Total: 8]